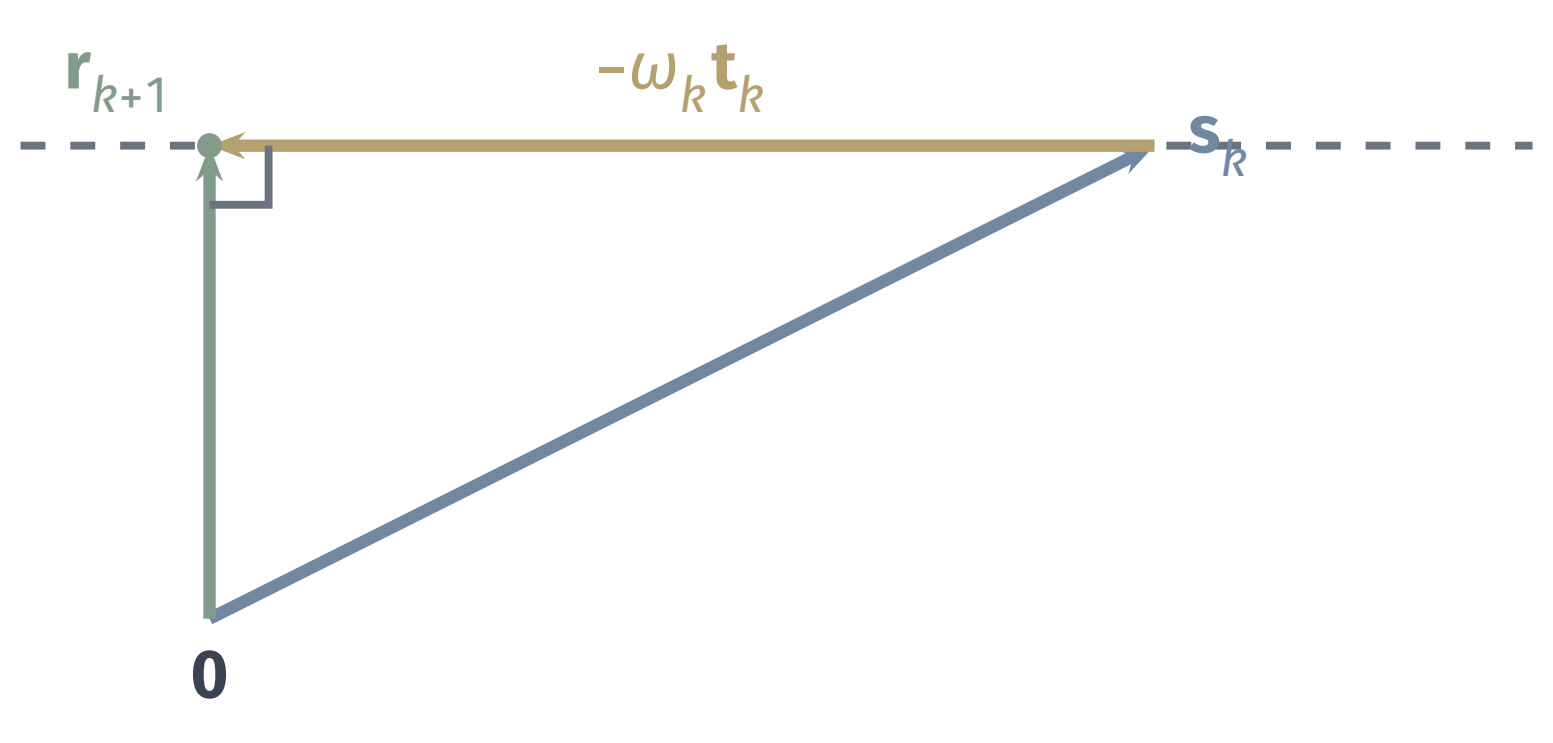
$\mathbf{r}_{k+1}$
$-\omega_k \mathbf{t}_k$
$\mathbf{s}_k$
0